Tertiary Infotech Academy Pte Ltd
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ACTIVITY WORKSHEETS

For

Agile Project Management for Business

TGS Ref No: TGS-2023018967

8 activities · one running case study: HarbourFront Logistics

Version v9.0 · 17 August 2026

ACTIVITY 1 WORKSHEET

Empathise with the Customer to Reframe a Failing Project

Agile Project Management for Business · TGS-2023018967 · Topic 1 · LO1 · 45 minutes

TEAM MEMBERS	**DATE**
TOOL https://alfredang.github.io/designthinking/	**ACTIVITY FOLDER** activities/activity-01-*/

THE SITUATION

HarbourFront Logistics spent 11 months building the CustomerConnect portal from a signed-off 96-page requirements document. Six months after launch, only 3 of 24 features are used weekly. The top customer complaint to the call centre is still "where is my shipment right now?" — a question the portal was supposed to answer. You are the newly appointed Agile project lead. Before writing a single new requirement, you must understand the customer.

Record your team's output below.

1. Persona name and role

2. SAYS (quote)

3. THINKS

4. DOES

5. FEELS

6. **Pains (at least 3)**

7. **Gains (at least 2)**

8. **REFRAMED PROBLEM STATEMENT (user + need + consequence, no solution)**

9. **What the ORIGINAL requirement optimised for instead**

10. **Top 3 features by customer value — state the BENEFIT, not the feature**

SELF-CHECK BEFORE YOU FINISH

- ☐ Your reframed problem statement names a specific user, a specific need and a specific consequence — and contains no solution or technology. Your top 3 features each state a customer benefit. If any 'feature' is a module name or a format, it is still a requirement, not a value statement — rewrite it.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.

ACTIVITY 2 WORKSHEET

Diagnose the Waterfall Failure with a Fishbone Analysis

Agile Project Management for Business · TGS-2023018967 · Topic 1 · LO1 · 45 minutes

TEAM MEMBERS ____________________	**DATE** ____________________
TOOL https://alfredang.github.io/fishbone/	**ACTIVITY FOLDER** activities/activity-02-*/

THE SITUATION

The HarbourFront management committee wants to know why CustomerConnect was 4 months late and 62% unused before it approves any new funding. The programme director's explanation was "the team underestimated". The CEO is not satisfied with that answer and has asked your team for a structured diagnosis by Friday.

Record your team's output below.

1. **Problem statement (the effect)**

2. **Causes — Requirements**

3. **Causes — Process**

4. **Causes — Customer Involvement**

5. **Causes — Governance**

6. **Causes — People**

7. **Causes — Technology**

8. **Count of causes tagged STRUCTURAL (S)**

9. **Count tagged BEHAVIOURAL (B)**

10. **Structural cause → the Agile practice that addresses it (one line each)**

11. **At least TWO causes Agile does NOT fix, and what would fix them**

12. **One-page recommendation to the committee (3–5 sentences)**

SELF-CHECK BEFORE YOU FINISH

- ☐ Your diagram carries at least 18 causes across all 6 categories, every cause is tagged S or B, and each structural cause is paired with a named Agile practice. Your recommendation honestly identifies at least two causes Agile will not fix — a diagnosis that concludes 'Agile solves everything' has not been done properly.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.

ACTIVITY 3 WORKSHEET

Build the Product Backlog and Run Sprint 1 Planning

Agile Project Management for Business · TGS-2023018967 · Topic 2 · LO2 · 60 minutes

TEAM MEMBERS	DATE
TOOL	**ACTIVITY FOLDER**
https://alfredang.github.io/scrum/	activities/activity-03-*/

THE SITUATION

HarbourFront's committee has approved a restart: 6 sprints of 2 weeks, one team of 7 people, and a hard demonstration to the top 5 customers in 12 weeks. Nothing from the old 96-page document is carried over unexamined. You are the team, with your trainer as Product Owner. Sprint 1 planning starts now.

Record your team's output below.

1. **Sprint 1 SPRINT GOAL (one business outcome, not a list)**

2. **Story 1 — text, points, Given/When/Then**

3. **Story 2 — text, points, Given/When/Then**

4. **Story 3 — text, points, Given/When/Then**

5. **Total stories in the product backlog (target 12+)**

6. **MoSCoW split — count of Must / Should / Could / Won't**

7. **Stories pulled into Sprint 1, and total points (target ~20)**

8. **Any story you SPLIT, and how you split it**

9. **Your team's DEFINITION OF DONE (checklist)**

10. **Confirm: does every Sprint 1 story serve the sprint goal? Which did you remove?**

SELF-CHECK BEFORE YOU FINISH

- ☐ Sprint 1 holds about 20 points, every story in it serves the one written sprint goal, no story exceeds 8 points, every story has Given/When/Then acceptance criteria, and the Definition of Done is visible to the whole team. A sprint whose stories serve four unrelated goals is a task list, not a sprint.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.

ACTIVITY 4 WORKSHEET

Clarify Agile Role Accountability with a RACI Matrix

Agile Project Management for Business · TGS-2023018967 · Topic 2 · LO2 · 45 minutes

TEAM MEMBERS

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DATE

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TOOL

https://alfredang.github.io/raci/

ACTIVITY FOLDER

activities/activity-04-*/

THE SITUATION

Two weeks into the restart, CustomerConnect has stalled in a familiar way. The old programme director is still approving every story change. The Scrum Master has been asked to produce a weekly percent-complete report. Two developers have escalated that they receive conflicting priorities from the Product Owner and from the operations manager. Nobody is behaving badly — the accountabilities were never made explicit.

Record your team's output below.

1. The 5 roles used as columns

2. Row with TWO Accountables in your first draft

3. Row with NO Accountable in your first draft

4. Which role was Consulted on too many rows, and what you changed to I

5. **How 'change the sprint scope mid-sprint' is assigned, and why**

6. **How 'report progress to the committee' changed**

7. **The specific duplicate A that caused the developers' escalation**

8. **The single change that most reduces the team's confusion, and why (2 sentences)**

SELF-CHECK BEFORE YOU FINISH

- ☐ Every one of the 12 rows has exactly one A. No role is C on more than about half the rows. The developers' conflicting-priorities escalation is traceable to a specific duplicate A in your first draft, and your corrected matrix removes it.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.

ACTIVITY 5 WORKSHEET

Execute Sprint 1 and Track It on the Scrum Board

Agile Project Management for Business · TGS-2023018967 · Topic 3 · LO3 · 60 minutes

TEAM MEMBERS	**DATE**
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TOOL	**ACTIVITY FOLDER**
https://alfredang.github.io/scrum/	activities/activity-05-*/

THE SITUATION

Sprint 1 of the CustomerConnect restart runs for 10 working days with a committed 20 points. Your team will simulate all 10 days in compressed time, holding a daily stand-up each 'day', moving cards, and hitting three real impediments the trainer injects: the carrier data feed returns stale timestamps on Day 3, a developer is pulled onto a production incident on Day 5, and the Product Owner requests a new story mid-sprint on Day 7.

Record your team's output below.

1. Sprint 1 committed points

2. WIP limit set on In Progress

3. Burndown — remaining points at Day 0,1,2,3,4,5,6,7,8,9,10

4. Impediment 1 (Day 3, stale timestamps) — decision and rationale

5. Impediment 2 (Day 5, capacity loss) — decision, and when you told the PO

6. **Impediment 3 (Day 7, new story) — displaced a story, or pushed to Sprint 2? Why?**

7. **ACTUAL VELOCITY (fully-Done points only)**

8. **Points CARRIED OVER (recorded separately)**

9. **Sprint Review feedback from the reviewing team**

10. **What your burndown shape tells you about your WIP**

SELF-CHECK BEFORE YOU FINISH

- ☐ Your burndown has 10 plotted points and an ideal line. Your velocity counts only fully-Done stories, with carryover recorded separately. All 3 impediments are logged with a decision and a rationale. If your burndown is a straight diagonal line, you have plotted the plan, not the sprint — plot what actually happened.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.

ACTIVITY 6 WORKSHEET

Run the Sprint Retrospective with 5 Whys Root-Cause Analysis

Agile Project Management for Business · TGS-2023018967 · Topic 3 · LO3 · 45 minutes

TEAM MEMBERS	**DATE**
____________________	____________________
TOOL	**ACTIVITY FOLDER**
https://alfredang.github.io/5whys/	activities/activity-06-*/

THE SITUATION

Sprint 1 finished 6 of 20 committed points short. In the review, one customer noted that the shipment status feature 'works but shows yesterday's data'. In the retrospective, the team's first instinct is to blame the carrier data feed. The Scrum Master pushes for a real root cause, because the same class of problem appeared twice in the old waterfall project.

Record your team's output below.

1. **The ONE problem chosen for analysis, and why that one**

2. **Why 1**

3. **Why 2**

4. **Why 3**

5. **Why 4**

6. **Why 5 (the root cause)**

7. **Backwards check — read the chain with 'because'. Does every link hold?**

8. Is the root cause something YOUR TEAM can change? (If no, keep asking why)

9. SMART action — owner, measure, and deadline

10. Story points allocated in the Sprint 2 backlog

11. Plus / Delta on the retrospective itself

SELF-CHECK BEFORE YOU FINISH

- ☐ Your chain reaches a process or system cause the team can change, not a person and not an external party. Reading it backwards with 'because' holds together at every link. Your SMART action has an owner, a measure and points in the Sprint 2 backlog. If your root cause is 'the carrier's fault', you stopped at Why 1.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.

ACTIVITY 7 WORKSHEET

Prioritise Defect Causes with a Pareto Chart

Agile Project Management for Business · TGS-2023018967 · Topic 3 · LO3 · 45 minutes

TEAM MEMBERS	**DATE**
TOOL https://alfredang.github.io/paretochart/	**ACTIVITY FOLDER** activities/activity-07-*/

THE SITUATION

Three sprints in, CustomerConnect has accumulated 120 logged defects. The team has capacity to attack roughly two root causes in Sprint 4, and the operations manager wants all 120 fixed. The Scrum Master has categorised every defect by cause and wants the team to decide with data rather than by whoever complains loudest.

Record your team's output below.

1. Total defects entered (must be 120)

2. Categories in descending order

3. Cumulative % after cause 1 / 2 / 3 / 4

4. The VITAL FEW — which causes, and the % they account for

5. **Link to Activity 6 — which root cause sits under the top two bars?**

6. **Sprint 4 target — numeric and verifiable**

7. **Your answer to the operations manager who wants all 120 fixed**

8. **What you are deliberately NOT fixing this sprint, and the cost of fixing it**

SELF-CHECK BEFORE YOU FINISH

- ☐ Your chart is sorted descending, the cumulative line reaches 100%, and your vital few are justified by the crossing point rather than chosen by intuition. Your Sprint 4 target is numeric and verifiable. If your recommendation is 'fix everything', you have not used the chart to make a decision.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.

ACTIVITY 8 WORKSHEET

Forecast the Release from Velocity and Read the Agile Metrics

Agile Project Management for Business · TGS-2023018967 · Topic 3 · LO3 · 45 minutes

TEAM MEMBERS	DATE
______________________	______________________
TOOL	**ACTIVITY FOLDER**
https://alfredang.github.io/scrum/	activities/activity-08-*/

THE SITUATION

The HarbourFront committee meets on Friday and wants one answer: when will the remaining CustomerConnect scope be delivered? The team has completed 4 sprints with velocities of 14, 18, 17 and 21 points. There are 186 points left in the backlog. The programme director wants a single date. The operations manager has separately asked why the support queue keeps growing despite the team 'going faster'.

Record your team's output below.

1. **Average velocity**

2. **Forecast — average / pessimistic / optimistic (in sprints)**

3. **Forecast in calendar weeks (range)**

4. **The assumptions your forecast depends on (list them)**

5. **Which CFD band is widening — the bottleneck**

6. **Cycle time trend from the control chart**

7. **Explain the growing support queue using Little's Law**

8. **Your recommended action (and why it is NOT 'increase velocity')**

9. **The one-paragraph answer to the committee**

SELF-CHECK BEFORE YOU FINISH

- ☐ Your answer is a range with assumptions, not a single date. You identify Testing as the bottleneck from the CFD and confirm it with the control chart. You explain the growing queue with Little's Law. If your recommendation is 'increase velocity', re-read the control chart — that is what caused the problem.
- ☐ Every team member can explain the output, not just the person who typed it.
- ☐ The artefact is exported or screenshotted into this worksheet.